

Algebra 1
Unit 12
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

For questions 1-3, identify whether the statement is true or false. If it is false, rewrite the statement to make it true.

1. Stem-and-leaf plots are useful to show large quantities of data. True or False
Stem-and-leaf plots are useful to show small quantities of data
2. Line plots can show the range of small quantities of data. True or False
3. Box plots show overall shape for large quantities of data. True or False
4. A survey of 20 high school students was conducted to determine average time in minutes to travel to school. Which graphical displays would best display the data?
Select all that apply.

- ☒ Box plot
- ☒ Line plot
- ☐ Histogram
- ☒ Stem-and-leaf plot

5. Explain your reasoning for question 4.
Answers will vary. Line and Stem-and-Leaf plots are great for displaying individual data values. All three are also great for showing the distribution and the shape for a small quantity of data.
6. The data of the 20 high school students are below.

{20, 8, 12, 31, 9, 9, 11, 4, 23, 17, 16, 10, 10, 7, 21, 19, 7, 18, 15, 10}

Explain which display would be better for the data, a stem-and-leaf plot or a histogram.
Answers may vary. A Stem-and-Leaf plot because the data can be grouped in 10's, individual values can be shown, as well as the shape of the distribution of the dataset.

Appliances-R-Us was analyzing their inventory before their next sale. The frequency table shows which appliances are in stock.

Frequency Table

Type of Appliance	Frequency
Microwave	22
Dishwasher	18
Oven	27
Refrigerator	13

Relative Frequency Table

Type of Appliance	Frequency
Rap	27.5%
Hip hop	22.5%
Rock	33.75%
Other	16.25%

7. In order to create a bar graph, the data from the ☐ relative frequency
☒ frequency table is needed.
8. In order to create a circle graph, the data from the ☒ relative frequency
☐ frequency table is needed.
9. Which of the follow are types of displays for univariate categorical data? *Select all that apply.*

- ☒ Line Plot
 - ☒ Circle Graph
 - ☐ Stem-and-leaf plot
 - ☒ Frequency Table
 - ☐ Box plot
 - ☒ Bar Graph
 - ☐ Histogram

10. Which of the follow are types of displays for univariate numerical data? *Select all that apply.*

- ☒ Line Plot
 - ☐ Circle Graph
 - ☒ Stem-and-leaf plot
 - ☐ Frequency Table
 - ☒ Box plot
 - ☐ Bar Graph
 - ☒ Histogram

Algebra 1
Unit 12
Lesson 2 Practice

Name Answer Key
Date _____
Period _____

Identify whether each type of data is categorical or numerical. Then identify if it is univariate or bivariate.

1. The number of visitors to the beach and the average daily temperature.

☐ Categorical
☒ Numerical

☐ Univariate
☒ Bivariate

2. Whether it is sunny, partially cloudy, cloudy, or raining at the beach.

☒ Categorical
☐ Numerical

☒ Univariate
☐ Bivariate

3. The number of visitors to the beach on a summer day.

☐ Categorical
☒ Numerical

☒ Univariate
☐ Bivariate

Use the following for questions 4-6.

A local high school wanted to collect data on the number of absences of students for the 2020-2021 school year. The data of a sample of 20 high school students is shown.

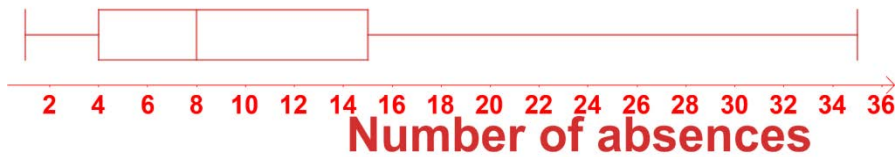
{11, 2, 15, 8, 8, 9, 20, 35, 1, 4, 23, 23, 7, 7, 6, 15, 3, 4, 1, 9}

4. This data set is ☐ categorical and ☒ univariate.
☒ numerical ☐ bivariate.

5. Which is the best graphical representation to display the data? Choose one.

☐ Scatterplot
☒ Box plot
☐ Line Plot
☐ Histogram
☐ Circle Graph
☐ Stem-and-leaf plot
☐ Bar Graph

6. Create a graphical representation for the data set display chosen in question 5.



The number of people in a household and the weekly grocery costs are shown in the table below of 10 different families.

Number of People in the Household	Weekly Grocery Cost (\$)
2	90
6	210
3	125
2	105
5	150
8	225
4	130
4	100
5	180
7	200

7. The data shown in the table is

☐ univariate.
☒ bivariate.

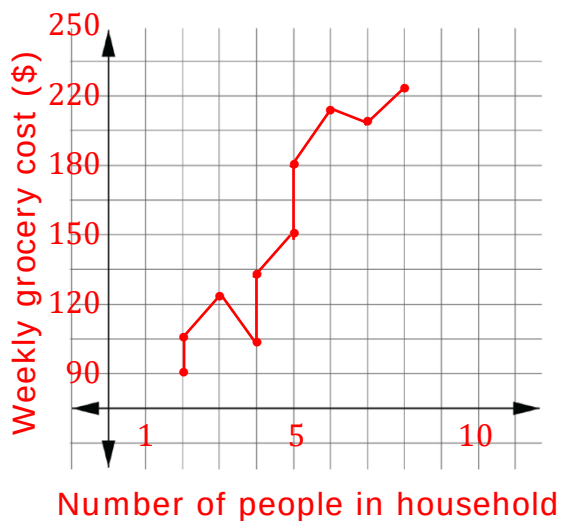
8. The data in the table should be displayed in a

☒ line plot.
☐ scatter plot.

9. Explain your reasoning from question 8.

Line plots allow for closer analysis from one point to the next.

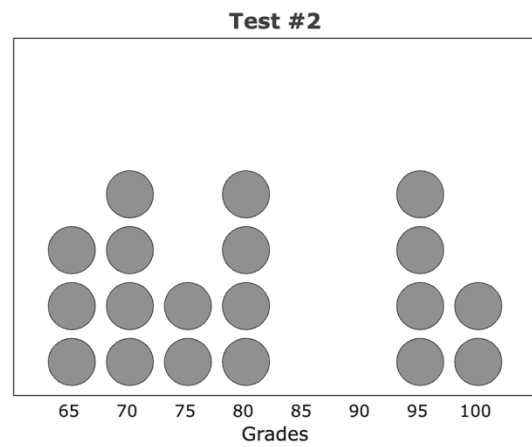
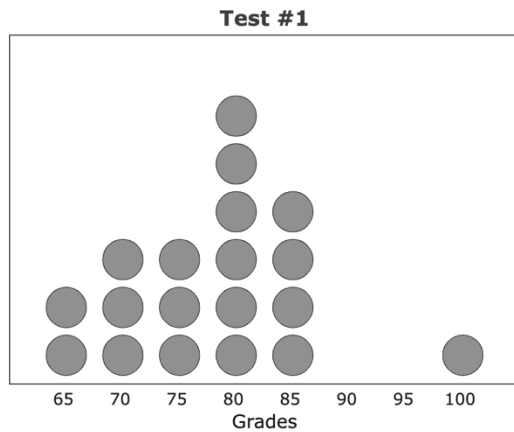
10. Create a display to represent the relationship between the number of people in a household and the weekly grocery costs.



Algebra 1
Unit 12
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

Two dot plots are shown, with results from two recent quizzes shown.



1. What is the mean of the scores for Test #1?

$$\frac{1485}{19} \approx 78.16$$

2. What is the median for the scores of Test #1?

80

3. Which measure of center is the most appropriate to use to describe the scores of Test #1? Explain how you know.

The data is skewed right due to an outlier so the median is the more appropriate.

4. Find the interquartile range of Test #1 and interpret it in the context of the problem.

$$85 - 70 = 15$$

5. What is the mean of the scores for Test #2?

$$\frac{1525}{19} \approx 80.26$$

6. What is the median of the scores for Test #2?

80

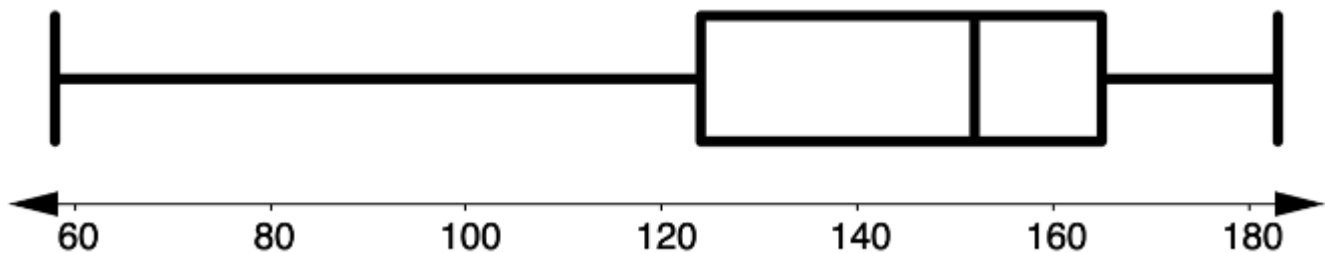
7. Which measure of center is most appropriate to use to describe the scores of Test #2?

Mean because the data has a fairly normal distribution.

8. Write a sentence comparing the results of Test #1 to Test #2 using measures of center and measures of spread.

The results in Test #1 are skewed, so it is best to use the median and IQR. The results in Test #2 have a fairly normal distribution so it is best to use the mean and standard deviation.

9. In the box plot shown, which measure of center would best be used to describe the data? How do you know?



The median because the 1st quartile is larger meaning the data is skewed.

10. A table of values for the heights of a group of tomato plants has a mean of 13 cm and a standard deviation of 0.2 cm. Interpret the variation in height.

The tomato plants are generally all the same height because there is a smaller standard deviation.

The heights of the plants range from 12.8–13.2.

Algebra 1
Unit 12
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

A circle graph is shown, with the results of a survey of a classroom of 135 students about which lunch entrée they prefer.

1. What is the most frequently chosen lunch entrée?

Pizza

2. What is the least frequently chosen lunch entrée?

Salad

3. How many students in the class chose tacos?

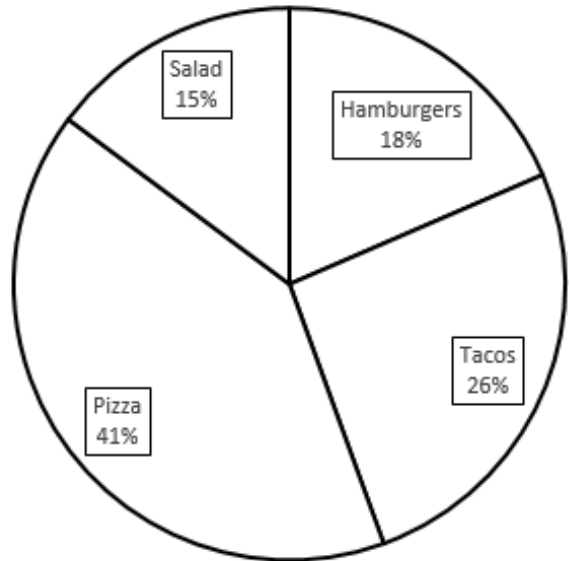
$0.26 \cdot 135 \approx 35$ students

4. How many students in the class chose salad?

$0.15 \cdot 135 \approx 20$ students

5. How many more students chose pizza over hamburgers?

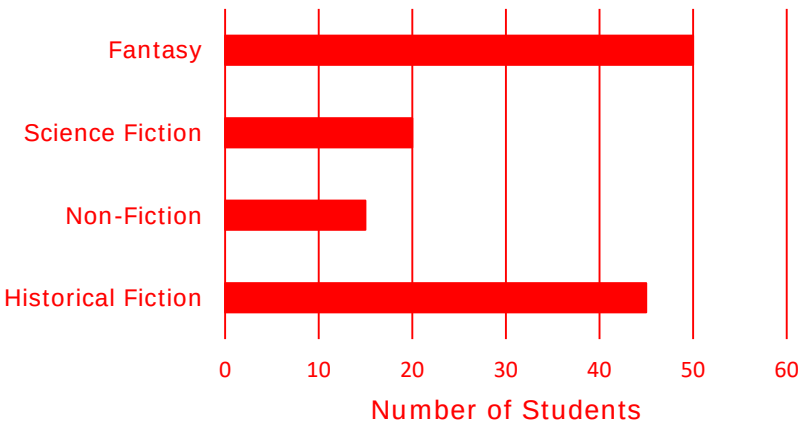
$(0.41 \cdot 135) - (0.18 \cdot 135)$
 $55.35 - 24.3 \approx 31$ students chose pizza over hamburgers



A table is shown with the results of a survey of a group of 130 students about their favorite type of book.

Favorite Book Genre	Number of Students
Historical Fiction	45
Non-Fiction	15
Science Fiction	20
Fantasy	50

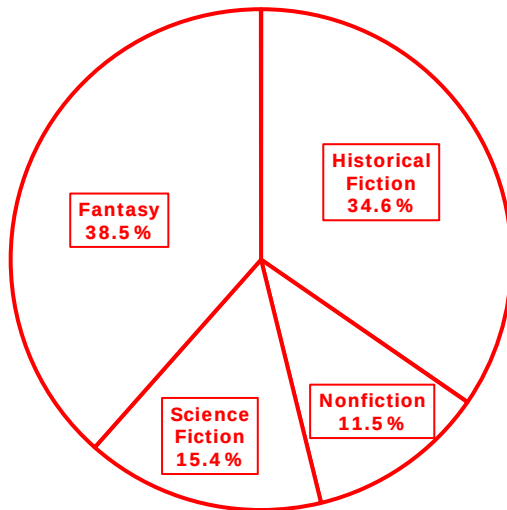
6. Create a bar graph showing the results of this survey.



7. Create a relative frequency table showing the results of this survey.

Favorite Book Genre	Number of Students	Relative Frequency
Historical Fiction	45	$\frac{45}{130} = 34.6\%$
Nonfiction	15	$\frac{15}{130} = 11.5\%$
Science Fiction	20	$\frac{20}{130} = 15.4\%$
Fantasy	50	$\frac{50}{130} = 38.5\%$
Total	130	$\frac{130}{130} = 100\%$

8. Create a circle graph showing the results of this survey



9. What percentage of students preferred either science fiction or historical fiction?
 $\text{Science Fiction} + \text{Historical Fiction} = 15.4\% + 34.6\% = 50\%$
10. What percentage of students did not prefer non-fiction?
 $100\% - 11.5\% = 88.5\%$

Algebra 1
Unit 12
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

A study shows that temperature has a negative correlation with hot chocolate sales.

1. What does it mean that there is a negative correlation?

As the temperature increases, hot chocolate sales decrease.

2. Does this mean that falling temperatures causes more hot chocolate to be sold?

No, correlation does not mean causation. There is just a relationship between the two variables.

3. What other factors might contribute to this association?

Hot chocolate may be more readily available and sold in falling temperatures but isn't necessarily offered or sold in warmer temperatures.

4. What other information would be helpful to determine whether falling temperatures causes hot chocolate sales to rise?

Prices, specials, availability

At a theme park, there is a positive correlation between the temperature increasing and the number of lunch time restaurant reservations.

5. What does it mean that there is a positive correlation?

As the temperature increases, the number of lunch reservations increase.

6. Does this mean that increased temperature leads to more lunch reservations being made?

No, just correlated but not necessarily the cause of increased reservations.

7. What other factors might contribute to this association?

Other weather factors, park attendance

At a water park, as rain increases, attendance decreases.

8. What kind of correlation does this association represent?

Negative

9. Is there evidence of a causal relationship?

No

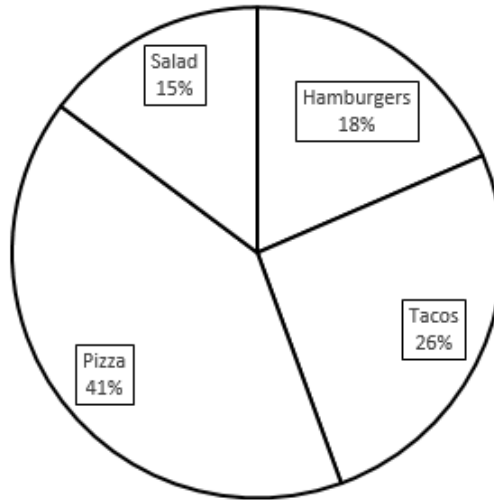
10. What other information would be helpful to evaluate the claim of a causal relationship?

It would need to be a controlled study or experiment.

Algebra 1
Unit 12
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

A circle graph is shown, with the results of a survey of middle school students about which lunch entrée they prefer, with a margin of error of 3%. There are 1250 students in the school.



1. Based on this survey, what is the estimated number of students who would prefer to have tacos?

$$0.26 \cdot 1250 = 325 \text{ students}$$

2. Based on this survey, what is the minimum number of students who would prefer to have hamburgers?

$$18\% - 3\% = 15\% \quad 0.15 \cdot 1250 \approx 187 \text{ students}$$

3. Based on this survey, what is the maximum number of students who would prefer to have salad?

$$15\% + 3\% = 18\% \quad 0.18 \cdot 1250 = 225 \text{ students}$$

4. Based on this survey, what is the range of students who would prefer to have tacos?

$$\begin{aligned} 26\% - 3\% &= 23\% & 0.23 \cdot 1250 &\approx 287 & \text{Approximately 287 to 363 students} \\ 26\% + 3\% &= 29\% & 0.29 \cdot 1250 &\approx 363 \end{aligned}$$

5. Based on this survey, what is the range of students who would prefer to have pizza?

$$\begin{aligned} 41\% - 3\% &= 38\% & 0.38 \cdot 1250 &= 475 & 475 \text{ to } 550 \text{ students} \\ 41\% + 3\% &= 44\% & 0.44 \cdot 1250 &= 550 \end{aligned}$$

A survey of parents indicated that 52% of students felt they have too much homework, with a margin of error of 2.1%.

6. Out of a group of 500 students, how many would likely feel that they have too much homework?

$$52 - 2.1 = 49.9 \quad 0.499 \cdot 500 \approx 249$$

$$52\% = 260 \text{ students}$$

$$52 + 2.1 = 54.1 \quad 0.541 \cdot 500 \approx 271$$

$$0.52 \cdot 500 = 260$$

$$\text{Range: } 249 \text{ to } 271 \text{ students}$$

7. Out of a group of 800 students, how many would likely not feel that they have too much homework?

$$100 - 52 = 48 \quad 0.48 \cdot 800 = 384$$

$$48\% = 384 \text{ students}$$

$$48 - 2.1 = 45.9 \quad 0.459 \cdot 800 \approx 367$$

$$\text{Range: } 367 \text{ to } 401 \text{ students}$$

$$48 + 2.1 = 50.1 \quad 0.501 \cdot 800 \approx 401$$

In a survey of 125 families, 24% reported that they eat out less than twice a week, 40% report they eat out twice a week, and 36% report they eat out more than twice a week. The survey has a margin of error of 1.5%.

8. What is the expected number of families that eat out less than twice a week?

$$0.24 \cdot 125 = 30$$

9. What is the range of families that eat out twice a week?

$$0.385 \cdot 125 \approx 48$$

$$48 \text{ to } 52$$

$$0.415 \cdot 125 \approx 52$$

10. What is the range of families that eat out less than twice a week?

$$0.225 \cdot 125 \approx 28$$

$$0.255 \cdot 125 \approx 32$$

$$28 \text{ to } 32$$